# Independent audit of Phase 2h: physical phase conventions and the Toda expansion

Quantum Boussinesq solitons project

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**Abstract**

The proposed Toda expansion remains worthwhile, but its numerical targets and the all-loop master-function interpretation require qualifications. **(d)** Independent phonon quadrature detects an orientation-dependent discrepancy between the Phase 2h master functional and the Phase 2g physical raw integrals. The claimed exchange antisymmetry of the raw phase is false in the regular right-moving sector. The saved edge-extrapolation alternatives do not resolve even the sign of the quoted subleading Toda coefficients. Conditional on the stated quasi-momentum shift, the classical Taylor remainder contains $(\log Y)^2/Y$, whose cancellation in the full phase is unproved. **(c)** We also correct the separated head-on kernel asymptotics and a displayed soliton profile. Existing calculations are left intact; this note supplies independent checks and a handoff for concurrent work.

## 1 Scope, conventions, and reproducibility

Labels follow the repository convention: (a) established in the literature, (b) heuristic in the literature, (c) derived or checked in this project, and (d) conjecture or research judgment. No new external literature claims are used here. All mathematical checks in this note are implemented in `phase2/scripts/audit_phase2h_foundations.py`, with results in `audit_phase2h_results.json`. The script uses 70-digit mpmath arithmetic and exact SymPy identities. Its JSON records SHA-256 hashes of the source documents, implementations, and saved data used, and it verifies that these inputs have not changed during execution. The checkout at the start was `0440ed6d345f7a8e94468b16209bc90b92bcdc6d`.

We retain the good-equation convention

$$u_{tt} = u_{xx} - u_{xxxx} - 3(u^2)_{xx}, \qquad u = 2(\log\tau)_{xx}, \qquad k = \sin\alpha, \quad v = \epsilon\cos\alpha.$$

The phase audit is in the regular region $0 < \alpha_j < \pi/3$; its directly integrated sample has $(\alpha_1, \alpha_2) = (0.3, 0.7)$. No new phase convention, regularisation, or definition of physical charge is selected. The Toda analysis reuses the saved Phase 2f data and does not claim a new continuum limit or a fresh high-resolution Bethe-ansatz solve.

**What survives.** The two existing Phase 2h scripts both finish with `ALL CHECKS PASSED`. Their algebraic coboundary tests and the Legendre-transform identity for the proposed collective-coordinate momentum are reproduced. **(c)** This audit distinguishes those identities from the physical identification of the quantities being compared.

## 2 The master functional is not yet the physical phase in all orientations

### 2.1 Independent phonon quadrature

For $\phi_j = \alpha + (2\pi/3, -2\pi/3, 0)_j$ and $c_j = (1, 1, -2)_j$, we implement the Phase 2e elementary phase independently:

$$\theta(q;\alpha,\epsilon) = -\sum_j c_j \left[\operatorname{sgn}(q)\arctan\frac{\sin\phi_j}{|q|} + \epsilon\arctan\frac{\sin\phi_j\cos\phi_j}{|q|\sqrt{1+q^2}+q^2+\sin^2\phi_j}\right],$$

$$I_{12}^{\mathrm{raw}} = \frac{1}{4\pi}\int_{\mathbb{R}\setminus\{0\}} \theta_1(q)\theta_2'(q)\,dq.$$

The phase tends to zero at each infinity, and the integral is split at zero. These are the existing raw-integral conventions, not a choice of physical exchange normalization. The independently implemented derivative and quadrature agree with the Phase 2g closed form to better than $10^{-60}$ at the four sampled orientations. **(c)** (`audit_phase2h_foundations.py`, check 1.)

| $\epsilon_1$ | $\epsilon_2$ | Direct raw integral | Phase 2h master | (master − raw)/$\pi$ |
|---|---|---|---|---|
| +1 | +1 | −1.10381243524 | −1.10381243524 | 0 |
| +1 | −1 | +0.49133439704 | +0.49133439704 | 0 |
| −1 | +1 | −0.49133439704 | −3.63292705063 | −1 |
| −1 | −1 | +1.10381243524 | −2.03778021835 | −1 |

The same closed-form mismatch is checked at two additional angle pairs. For a defect with $q_D = 0.4$, $\eta = +1$ and $\alpha = 0.3$, direct quadrature also agrees with Phase 2g, whereas the Phase 2h first-difference functional exceeds that raw soliton–defect integral by $\pi$ for $\epsilon = -1$ and agrees for $\epsilon = +1$. This example lies in the previously identified stable defect sector. **(c)**

**Interpretation.** The Phase 2h scripts compare a master-function difference to a newly defined ratio-kernel functional. Those comparisons can pass while the functional differs from the previously computed physical integral. The identification across orientations has therefore failed; the numerical algebraic coboundary identities themselves have not been disproved. **(c)**

### 2.2 Exchange constants cannot be dropped

For two right-moving regular solitons, the raw integral obeys

$$I_{12}^{\mathrm{raw}} + I_{21}^{\mathrm{raw}} = -\pi, \qquad I^{\mathrm{raw}}(\alpha,\alpha) = -\frac{\pi}{2}. \tag{1}$$

These statements are independently checked in `audit_phase2h_foundations.py`. **(c)** They agree with the integration-by-parts discussion in `oneloop_phase.tex`, but contradict the master report's unqualified claim of antisymmetry and the paragraph in `oneloop_defect_phase.tex` claiming that the raw phase vanishes on the diagonal. Adding $\pi/2$ gives an antisymmetric expression in this right-moving regular sector; it changes the bare triangle identities. This is a consistency problem to resolve, not a justification for selecting either normalization here. **(d)**

There is also no implication from oddness of a general kernel to exchange antisymmetry: the already-tested $K(z) = z/(1+z^2)$ yields $\mathcal{I}_{SS}(1,2) = \mathcal{I}_{SS}(2,1) = 2.29578918683$ at the same angles. The independent check reproduces that nonzero symmetric value. **(c)** Consequently, fusion within the algebraic representation does not by itself select a physical all-loop scattering amplitude. **(d)**

# 3 What the saved Toda data establish

## 3.1 The quoted subleading coefficients are not resolved

The coefficient used in the proposed next step is obtained from only two edge-extrapolated values:

$$\widehat{c}_1(r) = 32\,[R_0(16, r) - R_0(32, r)]. \tag{2}$$

This is the slope under an assumed linear dependence on $1/Y$, not an independent extraction of an asymptotic coefficient. **(c)** The saved `matching_analysis.json` already contains alternative edge fits. Applying (2) to each yields the following checked results (`audit_phase2h_foundations.py`, check 4):

| $r$ | Quadratic edge fit | Cubic edge fit | $h \log h$ edge fit | Propagated spread |
|---|---|---|---|---|
| 0.4 | −0.007080 | −0.007261 | +0.009148 | 0.018275 |
| 0.5 | −0.020991 | −0.021363 | +0.017819 | 0.045090 |
| 0.6 | −0.048219 | −0.048907 | +0.030317 | 0.093195 |

Here the last column is $32(s_{16} + s_{32})$, with $s_Y$ the recorded edge-fit spread. It is a sensitivity diagnostic, not a rigorous or statistical error bar. The sign change in the preceding columns uses a common alternative fit at both $Y$ values, so it does not depend on treating their errors as independent. **(c)** Better asymptotic control could eventually exclude that alternative, but the present data do not do so. **(d)**

## 3.2 The proposed leading limit remains a candidate

The central large-$Y$ intercepts lie within $1.3 \times 10^{-5}$ to $7.1 \times 10^{-5}$ of the proposed

$$c_0^*(r) = -\frac{4}{\pi}\big(\chi_2(r) - r\big).$$

This numerical proximity is reproduced. **(c)** However, the propagated edge-fit spreads are 0.00150, 0.00369, and 0.00761, respectively, more than fifty times those discrepancies. Agreement of central estimates does not establish accuracy at the displayed discrepancy scale. **(d)**

In `charge_and_frame_audit.py`, the statement

```
dPhi_loop_asymp = R0_target - dPhi_cl_asymp
```

builds the proposed answer into the loop asymptotic. The subsequent check is a numerical compatibility check, not an independent analytical derivation of the answer. **(c)** Claim 3.2 of `charge_frame_master.tex` asserts the corresponding loop expansion but does not derive it from the Bethe integral equation. This audit neither proves nor disproves $c_0^*(r)$; an independent edge-equation derivation is still needed. **(d)**

# 4 A logarithmic term that the next Toda calculation must track

Retain the Phase 2h classical edge function and its proposed shifts:

$$\Phi_{\rm cl}(u_1, u_2) = 2(u_1 - u_2)\log\frac{1 - \sqrt{u_2/u_1}}{1 + \sqrt{u_2/u_1}} + 4\sqrt{u_1 u_2},$$

$$(y_1, y_2) = (Y, r^2 Y), \qquad \delta u_1 = -\frac{\log Y}{2\pi} + c, \qquad \delta u_2 = \delta u_1 - \frac{\log r}{\pi}.$$

The shift formulas are inputs to this conditional calculation, not newly derived Bethe-ansatz results. Exact differentiation gives the quadratic Taylor term **(c)** (`audit_phase2h_foundations.py`, check 3):

$$Q = \frac{1}{2}\delta u^T D^2\Phi_{\rm cl}\,\delta u = \frac{(\delta u_2 - r^2\delta u_1)^2}{r(1 - r^2)Y}. \tag{3}$$

In particular,

$$Q = \frac{1-r^2}{4\pi^2 r}\frac{(\log Y)^2}{Y} + O\left(\frac{\log Y}{Y}\right), \qquad 0 < r < 1. \tag{4}$$

SymPy verifies the rank-one Hessian and this coefficient; an independent 70-digit Taylor comparison at $Y = 10^{12}$ verifies the quadratic term with relative error below $10^{-10}$. **(c)**

Thus the stated $O(Y^{-1})$ remainder after the first-order classical Taylor term is not a correct strict asymptotic bound under the proposed logarithmic shift. **(c)** This does *not* prove a surviving logarithm in the full residual: the intrinsic loop phase and further momentum corrections may cancel it. The next calculation must establish those cancellations before assuming a pure $c_1(r)/Y$ expansion. **(d)**

# 5 Additional checks: one correction and one positive control

## 5.1 The separated head-on kernels

Let $\alpha_1 = r\lambda$, $\alpha_2 = \lambda$, with directions $(+1, -1)$. Claim 4.3 of Phase 2h assigns a leading logarithm to each separated kernel. An independent expansion instead gives **(c)** (`audit_phase2h_foundations.py`, check 2)

$$I_{\chi_2}^{\text{head}} = \frac{9r}{\pi}\lambda^2 + O(\lambda^4), \tag{5}$$

$$I_{\ln}^{\text{head}} = \frac{r\lambda^2}{\pi}\left[-3\log(r\lambda^2) + 6\log 2 - \frac{3}{2}\log 3 - 12\right] + O(\lambda^4|\log\lambda|). \tag{6}$$

Their sum reproduces the existing total head-on formula, including its nonzero logarithmic term. The old script checked only that sum; its passing status did not verify the separately stated asymptotics. **(c)**

For a short derivation of (5), at zero amplitudes the three half-angle tangents and their derivatives are $w = (\sqrt{3}, -\sqrt{3}, 0)$ and $w' = (2, 2, 1/2)$. For $K_{\chi_2} = -\operatorname{Re}\chi_2/\pi$, the mixed derivative of $K_{\chi_2}(w_i w_j)$ is $-w_i' w_j'/[\pi(1 - w_i^2 w_j^2)]$. The weighted sum is $9/\pi$, verified exactly by SymPy. The sum is odd in each amplitude and analytic in a neighbourhood of zero, giving the displayed remainder. Independent numerical checks use $r = 0.25, 0.5, 0.75$ and $\lambda = 10^{-4}, 10^{-7}, 10^{-10}$. **(c)**

## 5.2 Charges and the profile

The identity $\Delta P_{\text{can}} = dL_1/dv$, using the proposed $L_1 = v\Delta P_{\text{mode}} - \Delta E_1^{\text{ren}}$, agrees with the existing closed form to better than $10^{-60}$ at the regular sampled points. **(c)** This is a positive algebraic control. It does not independently derive the renormalised effective action, or settle which full conserved charges enter soliton absorption and the Toda frame dictionary. Those physical identifications remain outside this audit. **(d)**

The displayed profile in Phase 2h also has a sign and amplitude error. From its stated $\tau = 1 + e^{k(x-vt)}$ convention one obtains

$$u = \frac{k^2}{2}\operatorname{sech}^2\frac{k(x-vt)}{2},$$

not $-k^2\operatorname{sech}^2(k(x-vt)/2)$. SymPy verifies the tau derivative. Substituting the quoted profile in the travelling-wave equation $u'' - k^2 u + 3u^2 = 0$ gives residual $9k^4/2$ at its centre. **(c)** This displayed error alone does not invalidate the implemented charge identities, which are checked separately.

## 6 Handoff, claims summary, and failures recorded

**Recommended division of work.** **(d)** The other investigation can proceed with the direct Toda edge expansion. It should treat $c_0^*$ as a candidate, retain logarithmic subleading terms, and avoid fitting to the quoted $c_1$ numbers. An all-loop physical kernel or ring quantisation should explicitly track orientation and zero-mode constants; the algebraic master functional cannot yet be substituted for the physical phase without that work. These recommendations select no new regularisation or normalization on the PI's behalf.

| Claim or test | Audit status | Evidence |
|---|---|---|
| Physical raw phase equals the master functional in all directions | Contradicted **(c)** | Check 1: independent quadrature; $\pi$ discrepancies |
| Raw phase is antisymmetric and vanishes on the diagonal | Contradicted **(c)** | Check 1; Eq. (1) |
| Algebraic coboundary tests | Existing tests pass **(c)** | Original master script rerun |
| Quoted $c_1$ values are established two-loop targets | Unsupported **(d)** | Check 4: saved fit alternatives reverse their signs **(c)** |
| Proposed exact $c_0^*$ limit | Open **(d)** | Numerical compatibility; independent derivation missing |
| First-order classical remainder is $O(1/Y)$ | Contradicted conditionally **(c)** | Check 3; Eqs. (3)–(4) |
| Separated head-on logarithms | Corrected **(c)** | Check 2; total asymptotic survives |
| Legendre-transform momentum formula | Confirmed conditionally **(c)** | Check 5; $< 10^{-60}$ residual |
| Displayed negative soliton profile | Contradicted **(c)** | Check 5; tau derivative and ODE residual |

**Failures recorded.**

1. The physical-identification, antisymmetry, diagonal-value, separated-head-on, strict-remainder, and displayed-profile claims fail the checks detailed above. These failures are retained explicitly as audit results; no tolerances were relaxed to make those claims pass.

2. The first audit run stalled while simplifying a Hessian written with nested square roots and was interrupted. Replacing $u_1 = s^2$, $u_2 = t^2$ and applying the exact chain rule makes the same symbolic Hessian check tractable; no mathematical assertion is weakened.

3. The tau-profile equality check initially rewrote only one side into exponentials, leaving a mixed exponential/hyperbolic expression unevaluated and failing the symbolic-zero assertion. Rewriting both sides consistently and applying exact rational cancellation proves the same identity. No numerical tolerance is involved.

**Verification scope.** The standalone audit and both original Phase 2h scripts were run. The full repository suite was not rerun: the audit changes no existing implementation, and does not claim to revalidate all earlier phases. `ALL CHECKS PASSED` in the new script means that its positive controls and reproductions of the recorded prior-claim failures passed. It does not mean the prior claims are now correct. The LaTeX note is built twice with `pdflatex` and its rendered pages are inspected.